

TABLEAU

Module 3 - Connecting Data Sources to Tableau

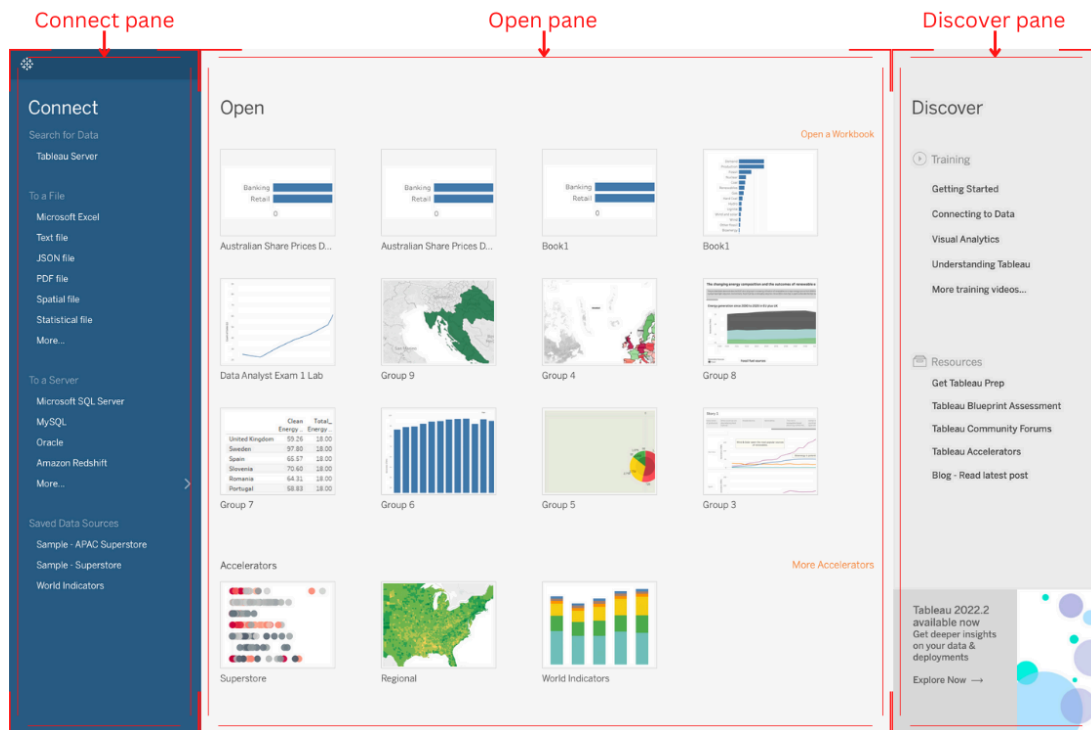
What You Will Learn in This Session

- | | |
|---------|--|
| Topic 1 | Live Connection vs Extract — What is the difference? |
| Topic 2 | Connecting Excel and CSV files to Tableau |
| Topic 3 | Connecting a MySQL Database to Tableau |
| Topic 4 | Relationships and Joins — combining multiple tables |

Beginner-Friendly • Real-Life Examples • Step-by-Step Instructions

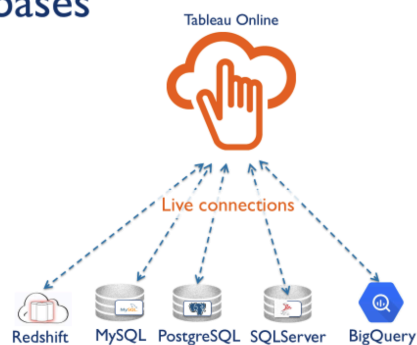
Topic 1 — Live Connection vs Data Extract

When you connect Tableau to any data source, you always get a choice: do you want a Live connection or an Extract? This is one of the first decisions you make. Let us understand both clearly.



1.1 What is a Live Connection?

Cloud Databases



Simple Definition

A Live Connection means Tableau is directly connected to your data source at all times. Every time you refresh a chart or open the dashboard, Tableau goes and fetches the latest data from the original source.

Real-life analogy: Think of it like watching a cricket match LIVE on Hotstar. You see every ball as it happens in real time. The moment a wicket falls, your screen updates. There is no delay — you are always seeing what is happening RIGHT NOW.

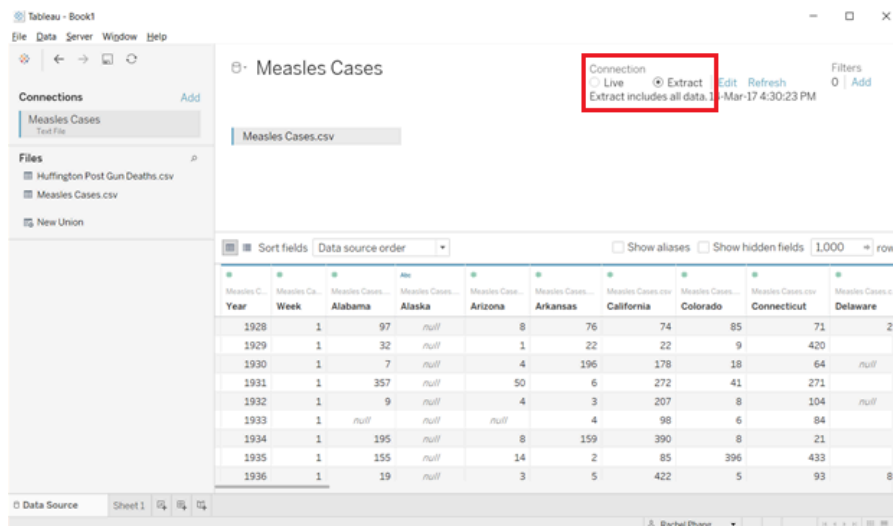
When does Live Connection make sense?

- Your data changes every few minutes — e.g. stock prices, website traffic, orders being placed
- You are connected to a fast database like MySQL, PostgreSQL, or Snowflake that can handle quick queries
- You need your dashboard to always show current numbers — e.g. a live sales monitoring screen

Drawbacks of Live Connection:

- If your database is slow or far away, your charts will load slowly
- If the database goes offline, your Tableau dashboard stops working
- More load is placed on the database server — costly for large organisations

1.2 What is a Data Extract?

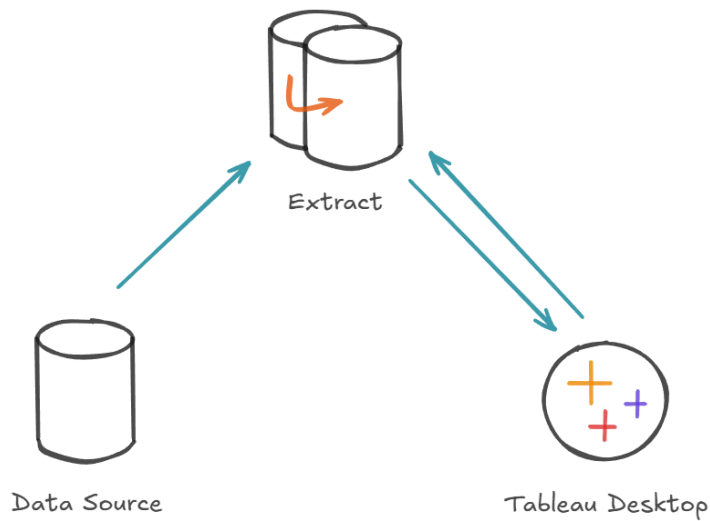


Simple Definition

An Extract means Tableau takes a snapshot copy of your data and saves it as its own file (called a .hyper file). Tableau then reads from this local copy instead of going to the original source every time.

Real-life analogy: Think of downloading a movie to watch offline. Once you download it, you can watch it anytime — on a flight, without internet, without buffering. The movie does not change after you download it, but you can always re-download a fresh copy whenever you want.

Extract Data Connection

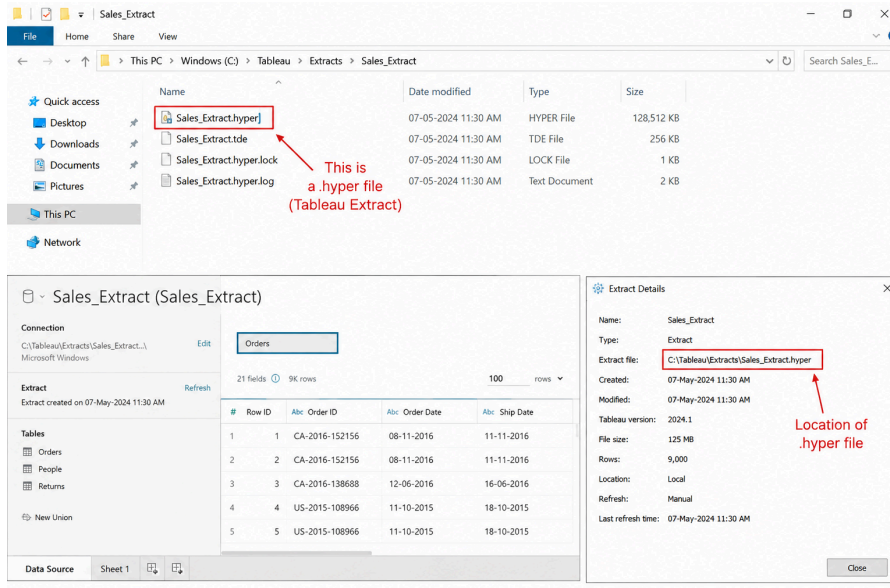


When does Extract make sense?

- Your data only updates once a day or once a week — e.g. end-of-day sales report from Excel
- You want fast performance — .hyper files are highly optimised and load very quickly



- You need to work offline, for example on a laptop without internet or VPN
- Your database is slow and you do not want to wait every time you open a chart



1.3 Side-by-Side Comparison

● Live Connection	● Data Extract (.hyper file)
Always shows latest data	Shows data from the time of last extract
Requires constant database connection	Works offline after extract is created
Speed depends on your database	Very fast — optimised Tableau engine
Good for real-time dashboards	Good for daily/weekly reporting
More load on the database server	No load on database while viewing
No setup needed — just connect	Must schedule extract refresh to stay updated

1.4 How to Choose — Decision Guide

Your Situation	Recommended Choice	Why
Live sales orders on website	Live Connection	<i>Data changes every minute</i>
Monthly expense report in Excel	Extract	<i>Data is static, speed matters</i>
Bank transaction database	Extract	<i>Security, offline access, speed</i>

Your Situation	Recommended Choice	Why
Hospital patient vitals monitor	Live Connection	<i>Real-time is critical</i>
Weekly inventory report	Extract	<i>Updated once a week is enough</i>
Social media analytics API	Extract	<i>Avoid repeated API calls</i>

1.5 How to Switch in Tableau

You can switch between Live and Extract at any time:

- **To switch to Extract:** On the Data Source tab, look at the top right — you will see two radio buttons: 'Live' and 'Extract'. Click 'Extract', then click 'Edit' to set filters if needed, then save.
- **To refresh an Extract:** Right-click the data source in the Data panel → Extract → Refresh. Tableau will go to the original source and create a fresh copy.
- **To switch back to Live:** On the Data Source tab, click the 'Live' radio button. Tableau reconnects directly.

Pro Tip for Beginners

When you are learning Tableau, always start with Extract. Your charts will load faster, you can work without the internet, and you will not accidentally overload a shared company database.

Topic 2 — Connecting Excel and CSV Files

Excel and CSV files are the most common data sources for beginners. Almost every organisation has data in spreadsheets. Tableau connects to them in just a few clicks.

2.1 What is an Excel File?

Simple Definition

An Excel file (.xlsx or .xls) is a spreadsheet file created in Microsoft Excel. It can have multiple sheets inside one file — like different pages. Each sheet can be a separate table of data.

Real-life example: A shopkeeper maintains a sales register in Excel. The file has 3 sheets: 'Orders', 'Products', and 'Customers'. Each sheet has rows of data with column headers in the first row.

2.2 What is a CSV File?

Simple Definition

A CSV file (Comma-Separated Values) is a plain text file where each row is a line and each column is separated by a comma. It is simpler than Excel — no colours, no formulas, just raw data.

What a CSV file looks like inside (you can open it in Notepad):

```
Order_ID, Customer_Name, Product, City, Sales
1001, Rahul Sharma, Laptop, Mumbai, 55000
1002, Priya Patel, Mobile Phone, Delhi, 18000
1003, Amit Verma, Headphones, Pune, 3500
```

Tableau reads this and automatically creates columns from the headers in line 1.

2.3 Excel vs CSV — Key Differences

Excel (.xlsx)	CSV (.csv)
Can have multiple sheets	Only one table of data per file
Supports formatting, colours, formulas	Plain text only — no formatting
Larger file size	Smaller file size — loads faster
Can have merged cells (causes issues)	Clean structure — no merged cells
Used widely in business	Used for data exports from systems
Tableau reads all sheets	Tableau reads it as a single table

2.4 Step-by-Step — Connecting an Excel File

1	Open Tableau Desktop. On the start screen, look at the left side under 'Connect'
2	Under 'To a File', click Microsoft Excel
3	A file browser opens. Navigate to your Excel file, select it, click Open
4	Tableau opens the Data Source screen. You will see your sheets listed on the LEFT side (like a drawer)

5	Drag your sheet name (e.g. 'Orders') to the white canvas area in the middle — this is called the 'drag sheets here' area
6	Tableau shows a preview of your data at the bottom — check that columns look correct
7	If your first row is headers (column names), Tableau auto-detects this. If not, check 'Field names are in first row'
8	Click the orange 'Sheet 1' tab at the bottom to start building your first chart!

Common Excel Problem — Merged Cells

If your Excel file has merged cells or empty rows at the top (like a company logo or title), Tableau may not read it correctly. Always keep your data starting from Row 1, Column A with clean column headers.

2.5 Step-by-Step — Connecting a CSV File

1	Open Tableau. On the start screen, under 'Connect → To a File', click Text File
2	Select your .csv file and click Open
3	Tableau automatically reads the first row as column headers
4	Check the Data Source preview at the bottom — all columns should be separated correctly
5	If columns look mixed up, check the separator (usually comma, but sometimes semicolon or tab)
6	Click Sheet 1 at the bottom to start working

2.6 Tips for Clean Excel and CSV Data

Before You Connect — Checklist

Following these rules will save you hours of frustration when connecting files to Tableau.

- **Row 1 must be your column headers** — e.g. 'Order_ID', 'Customer_Name', 'Sales'
- **No merged cells** — Tableau cannot understand merged cells; split them before connecting
- **No empty columns or rows in between data** — keep your data as one solid block
- **No total rows at the bottom** — rows like 'Grand Total: 5,00,000' confuse Tableau

- **Use consistent date formats** — either 'DD-MM-YYYY' or 'YYYY-MM-DD' throughout, not a mix
- **No special characters in column names** — avoid names like 'Sales (₹)' or '% Margin'; use 'Sales_INR' instead

Topic 3 — Connecting a MySQL Database

Files like Excel are fine for small datasets. But real businesses store millions of rows in databases. MySQL is one of the most popular databases in the world — used by Facebook, Twitter, Airbnb, and countless other companies.

3.1 What is a Database? What is MySQL?

Simple Definition — Database

A database is like a very large, organised filing system stored on a computer server. Instead of one Excel file, a database can hold hundreds of tables with millions of rows, and multiple people can access it at the same time.

Simple Definition — MySQL

MySQL is a type of database software (called a Relational Database Management System or RDBMS). It stores data in tables, just like Excel, but it is much more powerful, secure, and can handle enormous amounts of data.

Real-life analogy: If Excel is like a notebook where one person writes data, MySQL is like a library with thousands of books, managed by a librarian, where many people can search and read at the same time. You cannot just open the library yourself — you need a library card (username + password) to get in.

3.2 What You Need Before Connecting

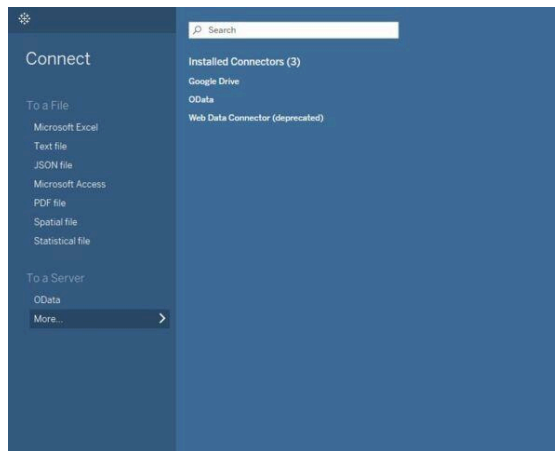
To connect Tableau to MySQL, you need 5 pieces of information. Your IT team or database administrator (DBA) will give you these:

What You Need	What It Means	Example Value
Server / Host	The address of the computer where MySQL is installed	localhost OR 192.168.1.10 OR db.mycompany.com
Port	The door number MySQL is listening on (default is 3306)	3306
Database Name	The specific database (library section) you want to access	sales_db OR ecommerce
Username	Your login ID for the database	tableau_user
Password	Your password for the database	***** (keep this secret!)

3.3 Step-by-Step — Connecting MySQL to Tableau

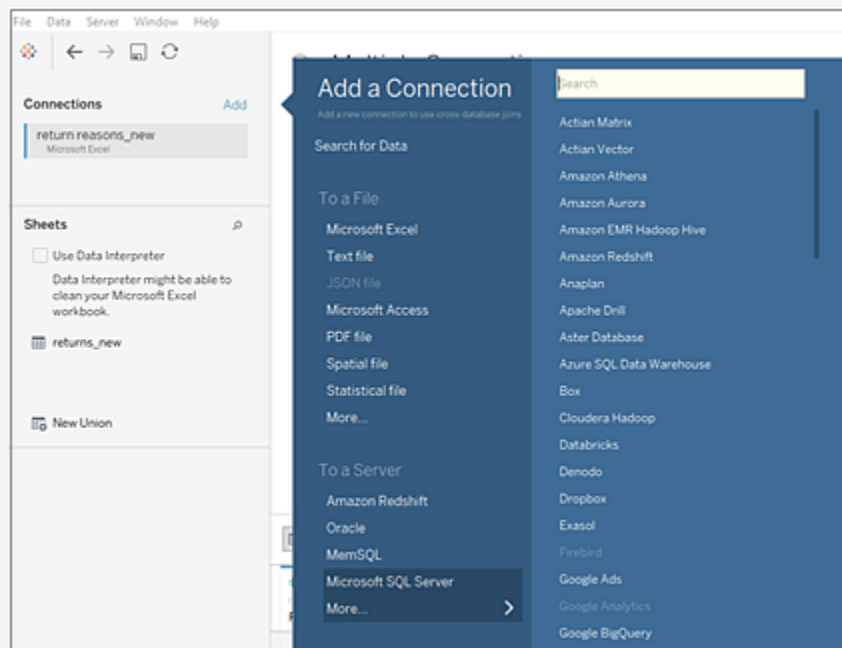
Open Tableau Desktop. On the start screen, look under 'Connect → To a Server'
If you do not see MySQL, click 'More...' and search for MySQL in the list

1



Click MySQL — a connection dialog box opens

2



Fill in the Server field: type the server address (e.g. localhost or the IP address your DBA gave you)

3

Fill in the Port field: usually 3306 (this is MySQL's default port — leave it as is unless told otherwise)

4

MySQL Connector/ODBC Data Source Configuration

MySQL Connector/ODBC

Connection Parameters

Data Source Name: mysql.servername

Description: My SQL Connector

☒ TCP/IP Server: mysql.servername.tsi.lan Port: 3306

☐ Named Pipe:

User: username

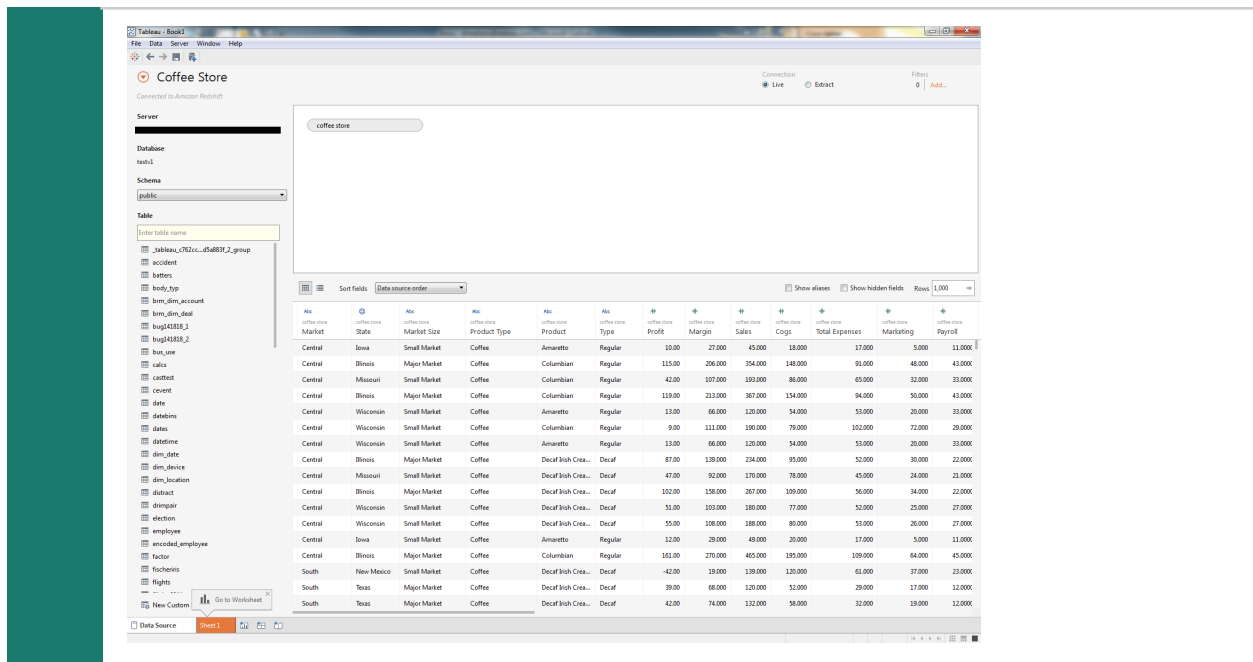
Password:

Database:

Test

Details >> OK Cancel Help

- 5 Fill in the Database field: type the name of the database you want to connect to
- 6 Fill in Username and Password: use the credentials your DBA provided
- 7 Click 'Sign In' — Tableau connects to MySQL (this may take a few seconds)
- 8 Once connected, you will see your database on the left panel
You can see all tables inside it — drag any table to the canvas to start working
- 9 Choose Live or Extract (see Topic 1). For large databases, Extract is recommended for better speed



What is 'localhost'?

'localhost' means the MySQL database is installed on the SAME computer you are working on. If you have installed MySQL on your own laptop for practice, use 'localhost' as the server address.

3.4 MySQL vs Excel — Key Differences

Excel File	MySQL Database
Stored on your computer as a file	Stored on a server, accessed over network
One user opens the file at a time	Hundreds of users can connect at once
Limited to ~10 lakh rows practically	Can store hundreds of crores of rows
No login needed (just open the file)	Requires username and password
Free to use	Requires MySQL installation and setup
Data can be accidentally edited	Role-based access — view only if needed

3.5 Troubleshooting Common Connection Errors

Error Message	What It Means and How to Fix It
Can't connect to MySQL server	Server address is wrong, OR MySQL is not running. Check the server name and ask your DBA if MySQL service is started.
Access denied for user	Wrong username or password. Double-check — passwords are case-sensitive. Contact your DBA.
Unknown database	The database name you typed does not exist. Ask your DBA for the correct database name.
Unable to connect to port 3306	Your firewall is blocking the connection. Ask IT to open port 3306 for your IP address.
Driver not found	MySQL connector driver is not installed. Download MySQL JDBC driver and add it to Tableau's driver folder.

Topic 4 — Relationships and Joins in Tableau

In real life, your data is almost never in one single table. Orders are in one table, customers in another, products in a third. To build meaningful reports, you need to combine these tables. This is what Joins and Relationships do.

4.1 Why Do We Need to Join Tables?

Real-life analogy: Think of an online shopping order. The order slip has the Order ID and Customer ID — but not the customer's name or address. The customer name is in a separate Customer Records book. To know WHO placed the order, you need to MATCH the Customer ID from the order slip with the Customer Records book. This matching is called a Join.

4.2 Our Practice Dataset — 3 Tables

We will use 3 tables from a small retail shop. These are the tables we will join throughout this topic:

Table 1: Orders

Order_ID	Customer_ID	Product_ID	Order_Date	Quantity	Sales (₹)
1001	C01	P01	05-Jan-2024	1	55,000
1002	C02	P02	10-Jan-2024	2	36,000
1003	C03	P03	12-Jan-2024	3	10,500
1004	C01	P01	18-Jan-2024	1	62,000
1005	C04	P04	22-Jan-2024	1	25,000

Table 2: Customers

Customer_ID	Customer_Name	City	Phone
C01	Rahul Sharma	Mumbai	9876543210
C02	Priya Patel	Delhi	9812345678
C03	Amit Verma	Pune	9823456789
C04	Kiran Kumar	Chennai	9834567890
C05	Deepa Nair	Bangalore	9845678901

Table 3: Products

Product_ID	Product_Name	Category	Cost (₹)
P01	Laptop	Electronics	42,000
P02	Mobile Phone	Electronics	12,000
P03	Headphones	Accessories	2,000
P04	Tablet	Electronics	18,000
P05	Keyboard	Accessories	1,500

Notice Something?

Customer C05 (Deepa Nair) exists in the Customers table but has NO order in the Orders table. And Product P05 (Keyboard) exists in Products but has not been ordered yet. Different types of joins will handle this differently — keep reading!

4.3 What is a Join?

Simple Definition

A Join is a way to combine rows from two tables by matching values in a common column. That common column is called the 'Join Key'. In our case, Customer_ID is the key that links the Orders table and the Customers table.

The screenshot shows a data tool interface with a sidebar on the left containing 'Connections' (Bookshop - Microsoft Excel) and 'Sheets' (Author, Award, Book, Checkouts, Edition, Info, New Union). The main area is titled 'Book+ (Bookshop)' and shows a join diagram where the 'Book' table is joined with the 'Award' table. Below the diagram, there are controls for 'Sort fields' (Data source order), 'Show aliases', 'Show hidden fields', and a row count of 18. The resulting data table is shown below:

Abc Award Title (Award)	Abc Award Award	# Award Year Won	Abc Book Book ID	Abc Book Title	Abc Book Auth ID
(im)Mortality	Young Reader's Choic...	2192	MM424	(im)Mortality	CW62

4.4 The Four Types of Joins

INNER JOIN — Only Matching Rows

Returns only the rows that have a MATCH in BOTH tables. If a row does not exist in the other table, it is excluded.

Real-life analogy: You have a list of students who registered for an exam, and a list of students who actually appeared. An Inner Join gives you only the students who BOTH registered AND appeared.

Result — Inner Join of Orders + Customers (matched on Customer_ID):

Order_ID	Customer_ID	Customer_Name	City	Sales (₹)	Included?
1001	C01	Rahul Sharma	Mumbai	55,000	✓ Yes — C01 exists in both
1002	C02	Priya Patel	Delhi	36,000	✓ Yes — C02 exists in both
1003	C03	Amit Verma	Pune	10,500	✓ Yes — C03 exists in both
1004	C01	Rahul Sharma	Mumbai	62,000	✓ Yes — C01 exists in both
1005	C04	Kiran Kumar	Chennai	25,000	✓ Yes — C04 exists in both
—	C05	Deepa Nair	Bangalore	—	✗ No — C05 has no orders

Deepa Nair is excluded because she has no orders. We get 5 rows.

LEFT JOIN — All rows from Left Table + Matching from Right

Returns ALL rows from the LEFT (first) table and only the matching rows from the RIGHT (second) table. If there is no match on the right, the right side shows NULL (empty).

Real-life analogy: A school has a list of all students. Some students have submitted their homework, some have not. A Left Join gives you ALL students — and shows the homework marks for those who submitted, and BLANK for those who did not.

Result — Left Join of Customers (Left) + Orders (Right):

Customer_ID	Customer_Name	City	Order_ID	Sales (₹)
C01	Rahul Sharma	Mumbai	1001, 1004	55,000 + 62,000
C02	Priya Patel	Delhi	1002	36,000

Customer_ID	Customer_Name	City	Order_ID	Sales (₹)
C03	Amit Verma	Pune	1003	10,500
C04	Kiran Kumar	Chennai	1005	25,000
C05	Deepa Nair	Bangalore	NULL	NULL ← No orders, still included!

Deepa Nair IS included with NULL values. All 5 customers appear.

RIGHT JOIN — All rows from Right Table + Matching from Left

Opposite of Left Join. Returns ALL rows from the RIGHT (second) table, and only matching rows from the LEFT table.

Real-life analogy: You have a list of job openings and a list of applicants. Right Join gives you ALL job openings — and shows applicant details where someone applied, and NULL where no one applied.

Result — Right Join of Orders (Left) + Products (Right):

Product_ID	Product_Name	Category	Order_ID	Orders placed?
P01	Laptop	Electronics	1001, 1004	✓ Ordered
P02	Mobile Phone	Electronics	1002	✓ Ordered
P03	Headphones	Accessories	1003	✓ Ordered
P04	Tablet	Electronics	1005	✓ Ordered
P05	Keyboard	Accessories	NULL	✗ Never ordered — still shown

Keyboard (P05) is included even though it has never been ordered. Useful to find unsold products.

FULL OUTER JOIN — All Rows from Both Tables

Returns ALL rows from BOTH tables. Where there is no match, the other side shows NULL. You get everything — matched and unmatched.

Real-life analogy: A hotel has a list of all rooms and a list of all guests. Full Outer Join gives you all rooms AND all guests — showing which guest is in which room, and which rooms are empty, and which guests have not checked in yet.

When to use Full Outer Join?

Use it when you cannot afford to lose any data from either table. Example: Auditing where you want to see both customers with no orders AND orders with no matching customer (data entry errors).

4.5 All 4 Joins — Visual Summary

Join Type	What it Returns	Left Table (Customers)	Right Table (Orders)
INNER JOIN	Only matching rows	<i>Only customers who placed orders (4 of 5)</i>	<i>Only orders with matching customer (all 5)</i>
LEFT JOIN	All left + matching right	<i>ALL 5 customers</i>	<i>Order data or NULL if no orders</i>
RIGHT JOIN	Matching left + all right	<i>Customer data or NULL if no match</i>	<i>ALL orders (5 orders)</i>
FULL OUTER JOIN	Everything from both tables	<i>ALL customers including unmatched</i>	<i>ALL orders including unmatched</i>

4.6 How to Create a Join in Tableau

1	Connect to your data source (Excel, MySQL, or any other source)
2	On the Data Source screen, drag the FIRST table (e.g. Orders) to the canvas
3	Drag the SECOND table (e.g. Customers) to the canvas — drag it RIGHT NEXT TO the first table You will see a JOIN icon appear (two overlapping circles) between them
4	Click the Join icon — a dialog box opens showing the join configuration
5	Select the Join Type: click Inner, Left, Right, or Full Outer
6	Set the Join Keys: on the left side select 'Customer_ID' (from Orders) On the right side select 'Customer_ID' (from Customers)
7	Tableau automatically previews the joined data at the bottom — check it looks correct
8	You can add more joins — drag a third table (Products) and connect it via Product_ID
9	Click Sheet 1 at the bottom to start building charts using the combined data

4.7 Relationships vs Joins — What is the Difference?

Important — Tableau has TWO ways to combine tables: Joins (the classic way) and Relationships (the newer, smarter way introduced in Tableau 2020.2)

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Joins (Classic Method)	Relationships (Modern Method)
Tables are physically merged into one flat table	Tables stay separate — Tableau queries them intelligently
Risk of duplicate rows if one table has many matches	No duplicate rows — handles one-to-many safely
You specify INNER/LEFT/RIGHT/FULL	No join type needed — Tableau decides automatically
Must be done on the Data Source screen	Done in the Data Model (canvas with circles, not Venn)
Better for simple, known table structures	Better for complex multi-table models
Available in all Tableau versions	Available from Tableau 2020.2 onwards

Recommendation for Beginners

Start with Joins — they are easier to understand and visualise. Once you are comfortable, explore Relationships for more complex projects. Both methods work, and many professionals use Joins for everyday work.

4.8 Creating a Relationship in Tableau

1	On the Data Source screen, drag your first table (Orders) to the canvas
2	Drag a second table (Customers) to the canvas — but this time drop it AWAY from the first table, not directly on top You will see it placed as a separate box connected by a line (not a Venn diagram)
3	Tableau will try to auto-detect the relationship key — it looks for columns with matching names
4	If it does not auto-detect, click the line between the two tables to open the relationship editor Set: Orders [Customer_ID] = Customers [Customer_ID]
5	Repeat for more tables — add Products and connect via Product_ID

Module 3 — Quick Reference Cheat Sheet

Topic 1 — Live vs Extract

Criteria	Live Connection	Data Extract
Data freshness	Always latest	As of last refresh
Speed	Depends on DB	Very fast
Works offline?	No	Yes
Best for	Real-time dashboards	Daily/weekly reports

Topic 2 — Excel and CSV

File Type	How to Connect	Key Tip
Excel (.xlsx)	Connect → To a File → Microsoft Excel	Remove merged cells before connecting
CSV (.csv)	Connect → To a File → Text File	Row 1 must be column headers

Topic 3 — MySQL

What You Need	Where to Find It
Server address	Your IT team / DBA (e.g. localhost or 192.168.1.10)
Port	Usually 3306 (MySQL default)
Database name	Ask your DBA (e.g. sales_db)
Username	Your database login ID
Password	Your database password — keep it safe!

Topic 4 — All 4 Join Types

Join Type	Returns	Use When
INNER JOIN	Only matching rows from both tables	You only want clean, complete records
LEFT JOIN	All left rows + matching right (NULL if none)	You want all records from main table
RIGHT JOIN	Matching left + all right rows (NULL if none)	You want all records from second table

Join Type	Returns	Use When
FULL OUTER JOIN	All rows from both tables	You cannot miss any record from either side

The Golden Rule of Joins

Always check your row count BEFORE and AFTER a join. If rows doubled or tripled, you likely have a one-to-many relationship causing duplicate rows. Switch to a Relationship instead, or fix your join keys.

End of Module 3 — Connecting Data Sources to Tableau

Practice: Connect the 3-table dataset (Orders, Customers, Products) using all 4 join types and observe how the row counts change!